

Hanaé Frotier de Bagneux

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OBJECTIVE

Second-year engineering student with a strong interest in power systems, renewable energy, and aerospace electrical systems, particularly embedded control and automation for satellite applications.

EDUCATION

CentraleSupélec x McGill

Electrical Engineering Boge – 2nd Year Student

Main Courses: Mathematics, Physics, Computer Science, Modelling, Automation

Saclay, France

September 2024 – July 2028

Lycée Français Victor Hugo

French General Baccalaureate – European Section, High Honours

Specialisations: Mathematics (advanced mathematics option), Physics & Chemistry, Economics & Social Sciences, Chinese

Marrakesh, Maroc

September 2022 – July 2024

PROFESSIONAL EXPERIENCE

ArianeGroup, Ground Systems Department

Electrical Ground Systems & Control-Command Intern

- Support the operational maintenance (MCO) of spacecraft electrical ground support equipment by developing and validating industrial control systems, ensuring the reliability and long-term performance of critical test infrastructures.

Vernon, France

June 2026 - July 2026

EDF (Electricité de France) - Research Assistant

Research Student, Cable Fault Detection

- Developing an interactive simulation tool for cable fault detection based on reflectometry techniques, analysing signal propagation and impedance discontinuities across power transmission lines.
- Identified and quantified energy losses induced by cable faults, contributing to infrastructure reliability analysis for energy distribution systems.

Fontainebleau, France

February 2026 – June 2026

Harel Mallac, NexEnergy

Solar Energy & Energy Performance Intern

- Performed electrical testing and fault diagnostics on photovoltaic modules, power cables, and three-phase transformers within a 2 MW solar power plant, analyzing transmission losses and system efficiency.
- Applied power engineering principles to evaluate plant performance, using data analysis to identify anomalies and propose technical optimization strategies for energy distribution and conversion systems.

Port Louis, Mauritius

July 2025 – August 2025

ENGINEERING PROJECTS & LEADERSHIP

EDF - Battery Systems Project

Battery Modelling

- Modelled a lithium-ion cell using a second-order Thévenin equivalent circuit, extracting RC parameters from pulse-response data and implementing an Extended Kalman Filter for real-time State of Charge estimation in an EV battery pack.

Fontainebleau, France

January 2026

Autonomous Line-Following Robot

Python Modelling Trainee | Control Systems

- Developed an autonomous trajectory-following robot based on Raspberry Pi
- Implemented PID control and multi-sensor perception (IR, ultrasonic, IMU) to ensure smooth navigation, slope detection, and safety functions

Paris Saclay, France

November 2025

PROJECTS & DISTINCTIONS

Morocco National Public Speaking Competition

Public Award for a speech on space debris and its impact on low-Earth orbit satellite reliability.

Meknes, Morocco

June 2024

Hack Your Future Competition

Developing a multifunctional smart cane to support people with disabilities or age-related loss of autonomy

Marrakesh, Morocco

May 2024

LANGUAGES & SKILLS

- Languages:** French (native), English (fluent) – IELTS: 8/9
- Technical Skills:** Python, MATLAB, Simulink, PSpice, Java, SQL, Microsoft Excel
- Electrical & Control Tools:** Circuit simulation (PSpice), State-space modelling, Feedback control design, Signal processing, Embedded systems, Power system analysis